

Leong Ko Ryan Jasper

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Singapore Citizen — eligible for H-1B1 (US-Singapore FTA), a no-lottery work visa similar to TN

EDUCATION

National University of Singapore

Aug 2023 – Aug 2026

Bachelor of Engineering (Honors), Major in Computer Engineering

Singapore

- GPA: 4.88/5.0 | Dean's List; Enrolled under the Engineering Scholars Program

Stanford University

Apr 2025 – Jun 2025

Visiting Student, Spring Quarter (Exchange)

Stanford, CA

EXPERIENCE

CV / ML Engineer

Jan 2025 – Present

ZeroshotData

San Francisco, CA

- Optimized ORB SLAM3 system for improved accuracy and real-time performance using computer vision algorithms and sensor fusion in C++ and Python
- Built EMF-based full body tracking system: trained and tested deep learning models achieving ± 1 cm positional and $\pm 5^\circ$ rotational precision
- Automated camera intrinsic calibration workflows using Kalibr, streamlining multi-camera setup and improving calibration accuracy for robotics perception systems
- Processed 100+ hours of gripper trajectory data for multi-modal dataset capture, enabling new training modalities for client robotic policies
- Integrated OAK-D-WIDE stereo cameras with DepthAI C++ bindings for real-time H.264 video processing and advanced computer vision pipelines
- Deployed ML/AI workloads on GCP using Kubernetes and containerized environments, reducing pipeline costs by 50% after first month

SWE Intern

May 2024 – Aug 2024

*A*STAR — National Metrology Centre*

Singapore

- Developed autonomous data-measurement web server using Python (PyVISA, Pandas, Dash) with data analysis and visualization for lab instrument data

SWE Intern

Apr 2023 – Jun 2023

HolyWally

Singapore

- Built full-stack dashboard using Python, SQL, and Superblocks with PostgreSQL, integrating REST APIs and third-party services for ~100 users

PROJECTS

Flair AI | *AI writing assistant; React, Tiptap, Twilio, Brainbase, Gemini, Tailwind*

2025

- **Winner (1st place, \$1,500)** — AI for Good Hackathon. Built end-to-end app with **automated essay grading** using NLP, live writing **metrics**, and grammar checks for ~400 students
- Integrated **Brainbase**, **Twilio**, and **Gemini** to deliver real-time **AI voice agent** with NLP, providing feedback in **English and Spanish**

WatchDocks | *Smart Surveillance; YOLOv8, Gemini API, Python, Flask*

2025

- Developed a real-time theft detection system using **YOLOv8**, fine-tuning the model to distinguish between normal usage and suspicious tampering
- Integrated **Gemini API** for context-aware analysis, reducing false positives by analyzing temporal patterns in video feeds

MedBox AI | *Medical Procedure Analysis; OpenCV, Chroma, Python*

2025

- Engineered a computer vision pipeline using **OpenCV** to automatically segment surgical procedures and detect medical tools in real-time
- Implemented a RAG system with **Chroma** vector database to retrieve relevant medical protocols and flag procedural deviations

SKILLS

ML/AI & Computer Vision: Python, PyTorch, TensorFlow, OpenCV, SLAM, Sensor Fusion, NLP, Deep Learning

Data Science: Pandas, NumPy, Scikit-learn, Data Analysis, Feature Engineering

Systems & Cloud: C/C++, ROS 2, GCP (GKE, Cloud Run, Pub/Sub, Artifact Registry, IAM), Kubernetes, Terraform, Docker, MLOps, DepthAI